

Optimization of Hybrid Biogas-Solar Energy Systems for Sustainable Power Generation in Mauritania

Sidahmed Sidi Habib*, Shuichi Torii*, Golla Suribabu[†], Anorgul Ismoilovnaa Ashirova[‡], Dilshodbek Mansurovich Allashkurov[§], Md Ashraful Islam[¶]

*Department of Advanced Mechanical System, Kumamoto University, Japan

[†]Civil Engineering Department, SRKR Engineering College (A), Bhimavaram A.P., India

[‡]Department of General Professional Sciences, Mamun University Khiva, Uzbekistan

[§]Department of Specialization, socio-humanities, and exact sciences, Tashkent State University of Economy, Uzbekistan

[¶]Dept of Electrical and Electronic Engineering, Green University of Bangladesh

Email: sidahmed23719@gmail.com, torii@mech.kumamoto-u.ac.jp, gsuribabu@srkrec.edu.in, anorgul76@gmail.com, allashkurovdilshod@gmail.com, mdashrafulislam3210@gmail.com

Abstract—This paper presents the optimization and performance analysis of hybrid biogas-PV systems for sustainable power generation in Rosso, Mauritania. The study employs HOMER software to model and optimize both on-grid and off-grid configurations, utilizing local data on solar irradiance, biomass availability, and load demand. The most optimal on-grid system configuration includes 801 kW of PV, a 100 kW biogas generator, and a 408 kW converter, achieving a levelized cost of energy (LCOE) of \$0.041/kWh and a net present cost (NPC) of \$1.89M. The off-grid system, with 177 kW of PV, a 100 kW biogas generator, 111 kWh of battery storage, and a 75.8 kW converter, yields an LCOE of \$0.128/kWh and an NPC of \$830,896. Advanced optimization algorithms, including the Grey Wolf Optimizer (GWO) and Whale Optimization Algorithm (WOA), were applied to further refine the results, with slight LCOE and system size improvements. Environmental impact assessments indicate significant reductions in greenhouse gas emissions, particularly with the off-grid system. The results highlight the feasibility and economic potential of hybrid biogas-PV systems in Mauritania and similar regions and emphasize the role of renewable energy in achieving sustainable and reliable power generation.

Index Terms—Hybrid systems, biogas, photovoltaic, renewable energy, HOMER, Mauritania, optimization, economic analysis.

I. INTRODUCTION

The reliance on fossil fuels as the primary source of energy has long been a dominant global trend. However, the adverse environmental impacts, including rising greenhouse gas emissions, air pollution, and climate change, have led to a global push toward more sustainable energy solutions [1]. The burning of fossil fuels contributes significantly to the increase in global temperatures, with scientists warning against exceeding a 1.5°C rise, as outlined in the 2015 Paris Agreement [2], [3]. This calls for urgent transitions to renewable energy sources to mitigate these effects. Renewable energy, particularly solar and biogas, offers a viable solution by harnessing naturally available resources to produce clean and sustainable electricity [4], [5], [6]. Numerous studies have explored the potential of hybrid renewable energy systems combining biogas and solar

photovoltaic (PV) technologies to address energy challenges in various regions [7], [8]. Ennemiri *et al.* evaluated the configuration and performance of a solar PV/biogas/battery hybrid system in Morocco, demonstrating that such systems are optimal in terms of cost savings and reducing greenhouse gas emissions when compared to systems relying solely on biogas or PV. The study highlighted that hybrid systems not only provide a reliable source of electricity but also contribute to environmental sustainability by reducing carbon emissions by approximately 40% [9], [10]. Similarly, Flavio *et al.* conducted a study on a solar PV/biogas/battery hybrid system for remote communities in Ghana. Using HOMER software for system modeling and optimization, their findings indicated that biogas could be effectively integrated with solar energy, resulting in a levelized cost of electricity (LCOE) of \$0.256 per kWh, which is lower compared to diesel-powered systems. This research underscores the economic viability of renewable hybrid systems for rural electrification [11]. In Nigeria, Jumare *et al.* assessed a grid-connected photovoltaic/wind/biogas system, showcasing the advantages of grid integration over off-grid configurations in terms of energy efficiency and cost reduction. The study highlighted that hybrid systems connected to the grid offer substantial savings in net present costs (NPC) and LCOE, making them more economically viable for long-term use [12]. A significant research gap exists in the optimization techniques applied in previous studies. Most studies on biogas-solar hybrid systems exclusively rely on HOMER software for simulation and optimization, overlooking the potential advantages of using advanced optimization algorithms. While HOMER offers a robust framework for system modeling, other optimization algorithms such as Grey Wolf Optimizer (GWO) and Whale Optimization Algorithm (WOA) have proven to be effective in solving complex optimization problems in various fields by providing superior exploration capabilities and convergence performance.

Mauritania, a country in West Africa, is heavily dependent

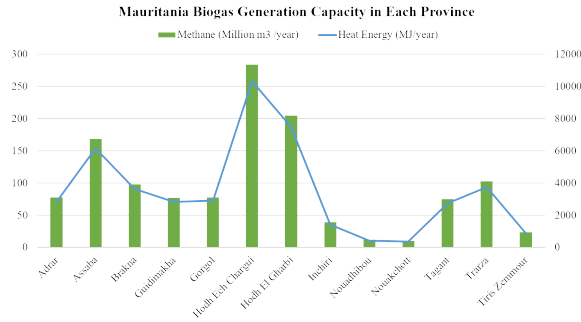


Fig. 1: Mauritania Biogas Generation Capacity in Each Province

on imported fossil fuels to meet its energy needs [13]. This dependency results in high energy costs, economic instability, and environmental degradation. Additionally, over 58.6% of the population lives in poverty, with limited access to reliable electricity, particularly in rural areas [6]. Mauritania has vast potential for renewable energy production, particularly from its abundant livestock resources, which offer a significant opportunity for biogas generation. With a livestock population exceeding 29.5 million, biogas can be produced from animal dung and crop residues, contributing to cleaner and more sustainable energy alternatives. Fig. 1 presents the Mauritania Biogas Generation Capacity in Each Province. Based on the identified gap in the literature and the requirement of Mauritania's remote area, the objective of this study is to assess the feasibility and optimize the integration of biogas and solar photovoltaic (PV) systems for electricity generation in Mauritania. This research aims to provide a comprehensive analysis using HOMER software along with advanced optimization algorithms - Grey Wolf Optimizer (GWO) and Whale Optimization Algorithm (WOA) - to identify the most cost-effective and reliable system configuration. This study seeks to offer a sustainable energy solution for regions with similar energy access challenges by addressing the research gap in optimization techniques and leveraging Mauritania's biogas resources.

II. METHODOLOGICAL FRAMEWORK FOR HYBRID BIOGAS-SOLAR ENERGY SYSTEM OPTIMIZATION

This study employs a comprehensive methodology to assess and optimize the performance of a hybrid biogas-solar photovoltaic (PV) system for sustainable electricity generation in Mauritania. The methodology encompasses multiple stages, including data collection, system modeling, simulation, and advanced optimization using both HOMER software and advanced optimization algorithms. This approach is designed to evaluate the proposed hybrid energy system's technical, economic, and environmental performance, providing valuable insights into its feasibility in Mauritania's energy landscape.

Advanced modeling techniques have been effectively used for solar energy integration purposes for optimal performance of the system.

A. Data collection

For this study, Rosso, Mauritania (coordinates: 16.5163°N, 15.8026°W) was selected as the target location for the hybrid biogas-solar energy system. For effective integration and optimal performance of the solar energy system, it's significant to model the radiation data in the solar collector [14]. Data on solar irradiance, biomass availability, and the local energy load profile were collected from surveys and relevant databases. The solar resource potential in Rosso in Fig. 3 shows an annual daily average radiation of 5.84 kWh/m²/day, with the highest values in May and June reaching up to 7.16 kWh/m²/day, while the lowest levels occur in December at 4 kWh/m²/day. Biomass availability, primarily sourced from livestock waste, remains relatively stable throughout the year, averaging 36.47 tonnes/day, providing a consistent biogas production potential. The highest biomass availability is recorded in January, April, August, and December with 37 tonnes/day, while May experiences the lowest with 35.56 tonnes/day. Fig. 4 depicts the amount of monthly average biomass resource available in Rosso. The daily load profile, reflecting the energy demand of the local population, peaks at 50 kW during the afternoon (15:00), gradually tapering off to 20 kW during the night and early morning hours. This load profile in Fig. 5 indicates the demand for consistent power throughout the day, particularly during the afternoon and evening hours. This comprehensive dataset, including solar radiation, biomass availability, and energy demand, forms the basis for simulating and optimizing the hybrid PV-biogas system in Rosso, Mauritania. Table I presents the cost information for the primary components used in the hybrid PV-biogas system, including photovoltaic (PV) panels, biogas generators, converters, and lithium-ion batteries. The PV panels, with a power capacity of 700 W, have a capital cost of \$400, a replacement cost of \$350, and an annual operation and maintenance (O&M) cost of \$5. The

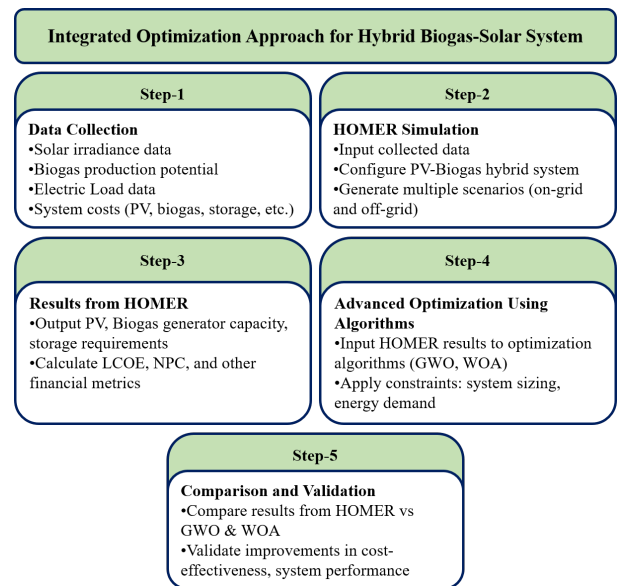


Fig. 2: Methodology framework for this optimization study.

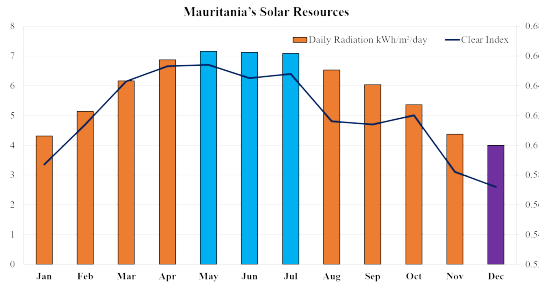


Fig. 3: Solar Resource Potential in Rosso, Mauritania.

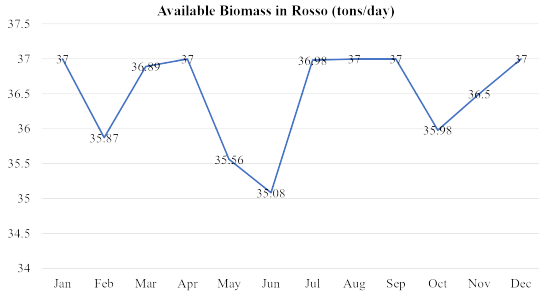


Fig. 4: Available Biomass in Rosso, Mauritania.

biogas generator, configured for auto-sizing, incurs a capital cost of \$500 per kW, a replacement cost of \$400 per kW, and an O&M cost of \$0.03 per operational hour. The converter, rated at 1000 W, has a capital cost of \$700 and a replacement cost of \$400, with no annual O&M costs. Lastly, the lithium-ion battery, with a capacity of 1 kWh, has a capital and replacement cost of \$550, alongside an annual O&M cost of \$10. These cost parameters are integral to the economic analysis and system optimization of the hybrid energy model for Rosso, Mauritania.

B. HOMER Simulation

The HOMER software was employed to model and optimize both on-grid and off-grid configurations of the hybrid PV-biogas energy system for Rosso, Mauritania. The simulation incorporated detailed input data, including solar irradiance, biomass availability, load profiles, and cost information for each system component. Two configurations were analyzed: (a) an on-grid system and (b) an off-grid system, as illustrated in Fig.6. In the on-grid configuration, the system is connected to the national electricity grid, allowing the sale of excess

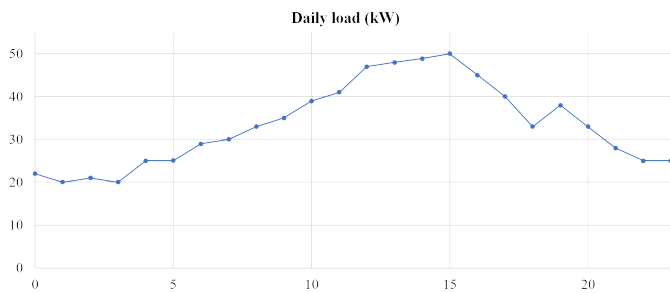


Fig. 5: Load profile for Rosso, Mauritania.

TABLE I: Cost Information of Components [15], [16].

Characteristics	PV Panel	Biogas	Converter	Battery
Model	EVO 6N SE6- 66HBD	Generic Genset	CSI- 1KTL1P- GI-FL	1kWh Li-Ion
Power (W)	700	Auto-size	1000	1 kWh
Capital Cost (\$)	400	500 \$/kW	700	550
Replacement Cost (\$)	350	400 \$/kW	400	550
O&M Cost (\$/year)	5	0.03 \$/op. hour	0	10

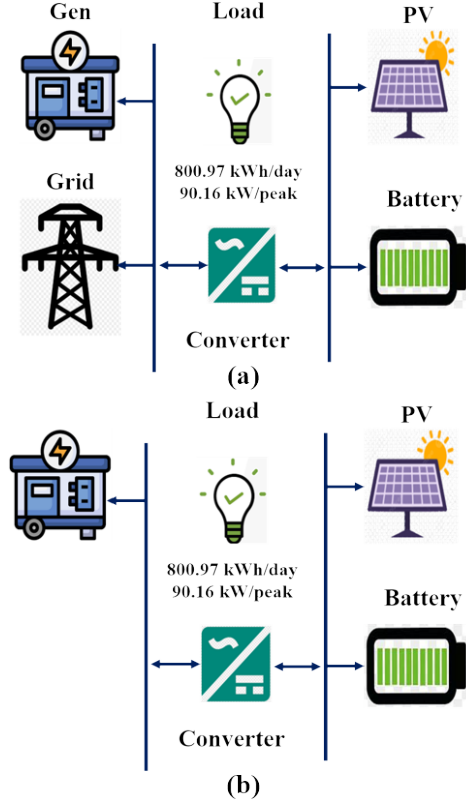


Fig. 6: (a) On-grid system configuration (b) Off-grid system configuration for HOMER optimization.

energy and reliance on the grid during low renewable energy production periods. The off-grid configuration, in contrast, relies solely on local energy generation, with battery storage playing a critical role in ensuring a consistent energy supply. Both configurations were optimized using HOMER's advanced algorithm to minimize the levelized cost of energy (LCOE), net present cost (NPC), and carbon emissions while maximizing system reliability. Key parameters, such as the size of PV panels, biogas generators, and storage capacities, were varied to identify the most cost-effective and efficient system for the selected location.

C. Optimization Using Algorithms

Two advanced algorithms were employed to enhance the hybrid PV-biogas system's optimization process: the Grey Wolf Optimizer (GWO) and the Whale Optimization Algorithm (WOA). As depicted in Fig. 7, the algorithm flowchart

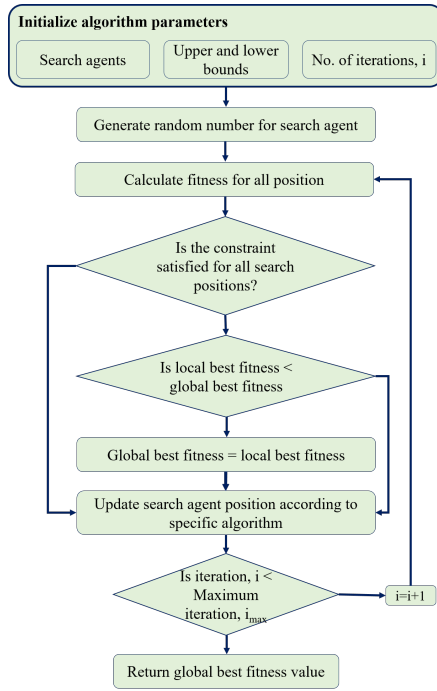


Fig. 7: Algorithm flowchart for system optimization.

outlines the sequential steps taken during the optimization phase. Initially, the best system configuration identified through HOMER simulations served as the input for both algorithms. The optimization aimed to minimize the levelized cost of energy (LCOE) and net present cost (NPC) while maximizing system efficiency. Both algorithms utilized the same constraints and equations for consistency. The GWO and WOA were chosen for their superior exploration capabilities, allowing for a comprehensive solution space search to identify optimal configurations. The results obtained from these algorithms were compared with those from HOMER to validate and potentially improve the system design for the specific conditions in Rosso, Mauritania. Fig. 7 presents the generalized flowchart for the used algorithms in this optimization process.

III. RESULT ANALYSIS

This study's result section presents the methodology employed to get an optimal Hybrid Biogas-Solar Energy system for green power generation. The evaluation examines vital parameters like the potential of energy generation, economic sustainability, and environmental conservation.

A. HOMER optimization result

The HOMER simulation provided insights into the optimal configurations for both on-grid and off-grid PV-biogas hybrid systems in Rosso, Mauritania. Key metrics such as energy production, LCOE, and NPC were analyzed to assess the system's technical and economic viability.

1) *Off-grid system result:* The optimal off-grid system configuration determined by HOMER Pro includes 177 kW of PV capacity, a 100-kW biogas generator, 111-kWh batteries, and a

75.8 kW converter. This system achieves an NPC of \$830,896, with an LCOE of \$0.128 per kWh and an annual operating cost of \$25,478. The initial capital investment required is \$265,219. The system has a 100% renewable energy fraction, meaning it relies entirely on renewable sources for electricity generation. The total annual electricity production amounts to 486,653 kWh, with 329,964 kWh generated by the PV system and 156,689 kWh by the biogas generator. Despite this, 172,051 kWh of electricity is excess, with no unmet electric load, indicating the system can fully meet the energy demand with surplus electricity.

2) *On-grid system result:* The optimal on-grid system configuration determined by HOMER Pro consists of 801 kW of PV capacity, a 100-kW biogas generator, and a 408-kW converter, with no battery storage. This setup achieves an NPC of \$1.89 million, an LCOE of \$0.041 per kWh, and an annual operating cost of \$120,886. The initial capital investment is \$793,314, with a renewable fraction of 100%. The system generates a total of 4,738,938 kWh annually, with 1,493,469 kWh from the PV array, 876,000 kWh from the biogas generator, and 2,369,469 kWh purchased from the grid. It meets the AC primary load of 292,354 kWh per year and sells 1,774,963 kWh of excess electricity back to the grid. Additionally, 239,452 kWh of electricity is considered excess, and there is no unmet load or capacity shortage, showcasing its high efficiency and ability to generate surplus energy.

3) *Economic result by HOMER optimization:* The economic analysis of the on-grid and off-grid hybrid PV-biogas systems reveals notable differences in financial metrics, as shown in Table IV. The on-grid system has a significantly lower LCOE of \$0.041 per kWh compared to the off-grid system's \$0.128 per kWh. This lower LCOE highlights the cost-effectiveness of integrating the grid with renewable sources. Additionally, the on-grid system has a shorter break-even period of 6.6 years, as opposed to 8.54 years for the off-grid system. Key financial indicators such as the IRR and ROI are also more favorable for the on-grid system, with IRR at 14% and ROI at 11%, compared to the off-grid system's IRR of 9.9% and ROI of 6.7%. The cost breakdown provided in Table V further illustrates the economic advantages of the on-grid system. Although the initial capital cost is higher for the on-

TABLE II: Off-grid system production summary.

Production Summary			
Component	PV	Biogas	Total
Production (kWh/year)	329,964	156,689	486,653
AC Primary Load	292,354		
Excess Electricity (kWh/year)	172,051		
Unmet Electric Load (kWh/year)	0		

TABLE III: On-grid system production summary

Production summary			
Component	PV	Biogas	Grid Purchases
Production (kWh/year)	1,493,469	876,000	2,369,469
AC Primary Load (kWh/year)	292,354		
Grid Sales (kWh/year)	1,774,963		
Excess Electricity (kWh/year)	239,452		
Unmet Electric Load (kWh/year)	0		

grid system at \$793,314, compared to \$265,219 for the off-grid system, the on-grid system benefits from significantly lower operational and maintenance costs, particularly due to the absence of battery storage. The total system costs, including capital, replacement, operation and maintenance (O&M), and salvage values, show that the off-grid system incurs higher replacement and O&M costs for components such as the battery and converter. In contrast, the on-grid system demonstrates more favorable financial outcomes, especially with a total cost offset by energy sold back to the grid, as reflected by the negative salvage values. These economic results underscore the financial viability of the on-grid system for sustainable power generation in Rosso, Mauritania.

B. Result by optimization algorithms

To further enhance the economic and system performance of the on-grid configuration, two well-known optimization algorithms, Grey Wolf Optimizer (GWO) and Whale Optimization Algorithm (WOA), were employed to compare their results with HOMER's simulation. The off-grid system was not simulated with these algorithms due to the on-grid system's superior financial and technical outcomes, as indicated by its lower LCOE and higher return on investment. Optimizing the on-grid system with alternative algorithms aims to explore the potential for further improving system performance beyond what HOMER offers. As shown in Table VI, the optimization results obtained through GWO and WOA provide slight improvements over the HOMER optimization. The WOA achieved the best results, with a PV capacity of 781 kW, a 98 kW biogas generator, and a 388 kW converter, yielding the lowest LCOE of \$0.036 per kWh, an NPC of \$1.77 million, and a payback period (PP) of 6.1 years. The GWO produced similar results, with a LCOE of \$0.038 per kWh, an NPC of \$1.81 million, and a payback period of 6.4 years. Both algorithms managed to slightly reduce system costs and improve financial metrics compared to HOMER, though the improvements are not substantial enough to entirely replace HOMER's results.

C. Environmental Impact assessment

The environmental performance of both the on-grid and off-grid systems was assessed by analyzing the greenhouse gas (GHG) emissions associated with each configuration. Table VII outlines the emissions data for various pollutants, including carbon dioxide (CO₂), carbon monoxide (CO), unburned hydrocarbons, particulate matter (PM), sulfur dioxide (SO₂), and nitrogen oxides (NO_x). The results show that the off-grid system significantly outperforms the on-grid system in

terms of environmental impact, primarily due to its complete reliance on renewable energy sources and the absence of grid dependence. The off-grid system emits only 10.3 kg/year of CO₂ compared to the on-grid system's 53.7 kg/year. Similarly, emissions of CO, NO_x, and particulate matter are also lower for the off-grid system. Notably, sulfur dioxide emissions are zero for both systems, further emphasizing the reduced environmental footprint of the biogas-PV hybrid systems. Although the on-grid system is more economically viable, the off-grid system offers superior environmental benefits, making it a more sustainable option for regions prioritizing reduced GHG emissions. These findings underscore the environmental advantages of renewable energy integration in Mauritania, where transitioning away from fossil fuels can substantially mitigate the country's carbon footprint.

IV. DISCUSSION

The findings from this study underscore the potential of hybrid biogas-PV systems to provide sustainable and reliable power solutions for regions such as Rosso, Mauritania. Both the on-grid and off-grid configurations achieved a 100% renewable energy fraction, demonstrating the feasibility of utilizing local biomass resources and abundant solar irradiance to meet energy demands. The economic comparison reveals a significant difference in Levelized Cost of Energy (LCOE) between the two systems: the on-grid system, with an LCOE of \$0.041/kWh, is more cost-effective and offers a shorter payback period of 6.6 years, making it highly viable for larger-scale, grid-connected projects. This cost advantage suggests that, where feasible, on-grid systems can be an economically sustainable option for enhancing energy infrastructure in Mauritania and similar regions, potentially attracting investment for grid-connected renewable energy expansions. The off-grid system, with an LCOE of \$0.128/kWh, provides an entirely renewable solution without relying on grid infrastructure, making it ideal for remote or underserved areas where grid access is limited or unreliable. Although the off-grid system's initial capital investment is lower, it requires a longer payback period of 8.54 years, highlighting the need for targeted incentives or financing mechanisms to make this model attractive for rural deployment. Given the economic and environmental implications, developing specific market adoption strategies, such as subsidies or blended financing, could facilitate broader adoption of both systems based on regional energy needs and access constraints. Applying optimization algorithms, including the Grey Wolf Optimizer (GWO) and Whale Optimization Algorithm (WOA), resulted in configurations with marginally lower LCOE and shorter payback periods compared to HOMER's baseline results. These algorithms showcase the potential for refining system performance and cost, particularly when financial constraints necessitate cost reduction. However, the observed improvements were incremental and did not surpass HOMER's results significantly, affirming HOMER's robustness in modeling hybrid energy systems for this context.

TABLE IV: Financial metric for both on-grid and off-grid systems.

Entry	Economic metrics	
	On-grid system	Off-grid system
LCOE (\$/kWh)	0.041	0.128
Break-even point (years)	6.6	8.54
Internal Rate of Return (%)	14	9.9
Return on Investment (%)	11	6.7

TABLE V: Cost summary of the optimized systems.

Cost Type	Capital (\$)		Replacement (\$)		O&M (\$)		Salvage (\$)		Total	
Components	Off-grid	On-grid	Off-grid	On-grid	Off-grid	On-grid	Off-grid	On-grid	Off-grid	On-grid
Biogas Generator	\$50,000	\$50,000	\$106,809	\$497,966	\$158,058	\$583,475	-\$1,427	-\$12,680	\$316,340	\$1,133,920
PV system	\$101,126	\$457,714	\$0.00	\$0.00	\$28,065	\$127,029	-\$11,688	-\$52,899	\$117,504	\$531,843
Battery	\$61,050	-	\$268,778	-	\$24,645	-	-\$28,819	-	\$325,653	-
Converter	\$53,042	\$285,600	\$26,363	\$141,945	\$0.00	\$0.00	-\$8,007	-\$43,112	\$71,398	\$384,433
Grid	-	\$0.00	-	\$0.00	-	-\$3,940,816	-	\$0.00	-	-\$3,940,816
System	\$265,219	\$793,314	\$401,949	\$639,911	\$210,768	-\$3,230,312	\$2,900	-\$108,692	\$49,940	-\$1,890,619

TABLE VI: Comparison of HOMER optimization results with algorithm results.

Optimization Algorithm	PV (kW)	BG (kW)	Battery	Converter (kW)	LCOE (\$/kW)	NPC (\$)	PP (years)
HOMER	801	100	-	408	0.041	1.89M	6.6
GWO	793	98	2	399	0.038	1.81M	6.4
WOA	781	98	-	388	0.036	1.77M	6.1

TABLE VII: GHG data for optimized systems.

Pollutants Name	Emissions (kg/year)	
	Off-grid System	On-grid system
Carbon Dioxide	10.3	53.7
Carbon Monoxide	1.08	5.63
Unburned Hydrocarbons	0.0470	0.246
Particulate Matter	0.00653	0.0341
Sulfur Dioxide	0	0
Nitrogen Oxides	1.01	5.29

V. CONCLUSION

This study successfully demonstrated the feasibility of hybrid biogas-PV systems for sustainable electricity generation in Mauritania, with a focus on the city of Rosso. Both the on-grid and off-grid configurations were optimized using HOMER software and further refined with advanced optimization algorithms, yielding highly efficient, renewable energy systems that address both economic and environmental challenges. The on-grid system emerged as the most economically viable option, with a low LCOE and faster payback period, while the off-grid system proved superior in terms of environmental impact, with significantly lower greenhouse gas emissions. The results suggest that hybrid biogas-PV systems can be adapted to similar regions, especially where biogas resources from livestock and agricultural waste are abundant, and solar irradiance is high. Future work could explore the integration of energy storage technologies and further refine the optimization process using other algorithms. Additionally, policy interventions aimed at financing renewable energy infrastructure will be critical for the widespread implementation of such systems in Mauritania and similar developing regions. This study underscores the significant role of renewable energy in addressing the dual challenges of energy access and environmental sustainability.

REFERENCES

- [1] M. A. Islam, A. A. Mamun, E. U. Khaled, A. Hasan, A. Islam, Z. U. Islam, and M. M. N. Ali, "Techno-economic assessment of electric bus for student transportation services: A case study," in *2023 5th International Conference on Sustainable Technologies for Industry 5.0 (STI)*, 2023, pp. 1–6.
- [2] M. A. Islam, M. N. Ali, A. J. Nahian, M. H. Maruf, M. S. H. Lipu, and A. Shihavuddin, "Prospect of electric boat in dhaka: an alternate way of transportation," in *2022 4th International Conference on Sustainable Technologies for Industry 4.0 (STI)*. IEEE, 2022, pp. 1–6.
- [3] M. A. Islam, M. N. Ali, A. Al Mamun, M. S. Hossain, M. H. Maruf, and A. Shihavuddin, "Optimizing energy solutions: A techno-economic analysis of solar-wind hybrid power generation in the coastal regions of bangladesh," *Energy Conversion and Management: X*, p. 100605, 2024.
- [4] S. Yousaf, I. Shafi, S. Din, A. Paul, and J. Ahmad, "A big data analytical framework for analyzing solar energy receptors using evolutionary computing approach," *Journal of Ambient Intelligence and Humanized Computing*, vol. 10, pp. 4071–4083, 2019.
- [5] M. J. B. Kabeyi and O. A. Olanrewaju, "Technologies for biogas to electricity conversion," *Energy Reports*, vol. 8, pp. 774–786, 2022.
- [6] S. Sidi Habib and S. Torii, "Biogas as alternative to liquefied petroleum gas in mauritania: An integrated future approach for energy sustainability and socio-economic development," *Clean Technologies*, vol. 6, no. 2, pp. 453–470, 2024.
- [7] I. B. Benitez and L. A. M. Danao, "Techno-economic evaluation of horizontal axis tidal stream turbines in san bernardino strait, philippines," in *2023 IEEE/IAS Industrial and Commercial Power System Asia (I&CPS Asia)*. IEEE, 2023, pp. 1485–1489.
- [8] I. B. Benitez, J. A. Ibañez, C. I. D. Lumabad, J. M. Cañete, and J. A. Principe, "Day-ahead hourly solar photovoltaic output forecasting using sarimax, long short-term memory, and extreme gradient boosting: Case of the philippines," *Energies*, vol. 16, no. 23, p. 7823, 2023.
- [9] N. Ennemiri, A. Berrada, A. Emrani, J. Abdelmajid, and R. El Mrabet, "Optimization of an off-grid pv/biogas/battery hybrid energy system for electrification: A case study in a commercial platform in morocco," *Energy Conversion and Management: X*, vol. 21, p. 100508, 2024.
- [10] J. A. Ibañez, I. B. Benitez, J. M. Cañete, J. C. Magadia, and J. A. Principe, "Accuracy assessment of satellite-based and reanalysis solar irradiance data for solar pv output forecasting using sarimax," *Journal of Renewable and Sustainable Energy*, vol. 15, no. 6, 2023.
- [11] F. Odoi-Yorke, S. Abaase, M. Zebilila, and L. Atepor, "Feasibility analysis of solar pv/biogas hybrid energy system for rural electrification in ghana," *Cogent Engineering*, vol. 9, no. 1, p. 2034376, 2022.
- [12] I. A. Jumare, R. Bhandari, and A. Zerga, "Assessment of a decentralized grid-connected photovoltaic (pv)/wind/biogas hybrid power system in northern nigeria," *Energy, Sustainability and Society*, vol. 10, pp. 1–25, 2020.
- [13] K. Ntaghry, A. Thiam, S. M. S. Habib, K. Faye, and M. Faye, "Evaluation of suitable sites for concentrated solar power desalination systems: case study of mauritania," *Environmental Research Communications*, vol. 6, no. 8, p. 085020, 2024.
- [14] S. Yousaf, I. Shafi, S. U. Butt, and M. Farooq, "An analytical model for pressure loss coefficients in tee junction of a solar collector based upon discharge ratio and reynolds number," in *ASTFE Digital Library*. Begel House Inc., 2019.
- [15] "Assessing the feasibility and quality performance of a renewable energy-based hybrid microgrid for electrification of remote communities," *Energy Conversion and Management: X*, vol. 23, p. 100674, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2590174524001521>
- [16] M. A. Islam, A. Al Mamun, M. N. Ali, R. H. Ashique, A. Hasan, M. M. Hoque, M. H. Maruf, M. A. Al Mansur, and A. Shihavuddin, "Integrating pv-based energy production utilizing the existing infrastructure of mrt-6 at dhaka, bangladesh," *Heliyon*, vol. 10, no. 2, 2024.